A

Review on

"A Hybrid Approach for Image Enhancement and Visibility Restoration Under Atmospheric Degradation"

Under the

Department of

ELECTRONICS AND COMMUNICATION ENGINEERING

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ABSTRACT

- **Sand-dust** refers to a mixture of **fine sand and dust particles** that are **suspended in the air** due to wind or storms. These particles **scatter and absorb light**, reducing visibility and causing hazy or yellowish images.
- Outdoor images captured in fog, smoke, or sand-dust suffer from color distortion, low contrast, and poor visibility.
- A fast adaptive enhancement algorithm is proposed for various degradation conditions.
- Uses color channel compensation to restore attenuated colors (blue for sand-dust, gray for fog/smoke).
- Fog, smoke, and dust reduce image clarity, contrast, and details.
- Dark Channel Prior (DCP) is used to estimate haze and restore visibility.
- Transmission map and atmospheric light help recover the clear image.
- Results show improved quality using PSNR and SSIM metrics.

INTRODUCTION

- Images captured in foggy or hazy weather often suffer from low contrast and poor visibility.
- Haze caused by atmospheric particles reduces clarity and hides scene details.
- Image dehazing techniques help restore the true appearance of the scene.
- The Dark Channel Prior method is used to estimate and remove haze effects.
- After dehazing, contrast enhancement improves brightness and visual quality.
- This approach produces clearer, more natural images with improved visibility and detail.

Fig.1: Low contrast images

LITERATURE SURVEY

	Year	Reference	motive	Author
1	2025	"A novel intensity-corrected blue channel compensation and edge-preserving contrast enhancement using laplace filter and sigmoid function for sand-dust image enhancement." IEEE Access 13 (2025)".	Fast haze removal using adaptive gain.	Tung, Tzu-Chia, et al.
2	2024	"Image Enhancement Thanks to Negative Grey Levels in the Logarithmic Image Processing Framework." Sensors 24.15 (2024).	Enhance images via negative gray levels.	Jourlin, Michel.
3	2024	"EV-fusion: A novel infrared and low-light color visible image fusion network integrating unsupervised visible image enhancement." IEEE Sensors Journal 24.4 (2024).	Enhancing visibility through advanced multimodal image fusion.	Zhang, Xin, et al

EXISTING METHODOLOGY

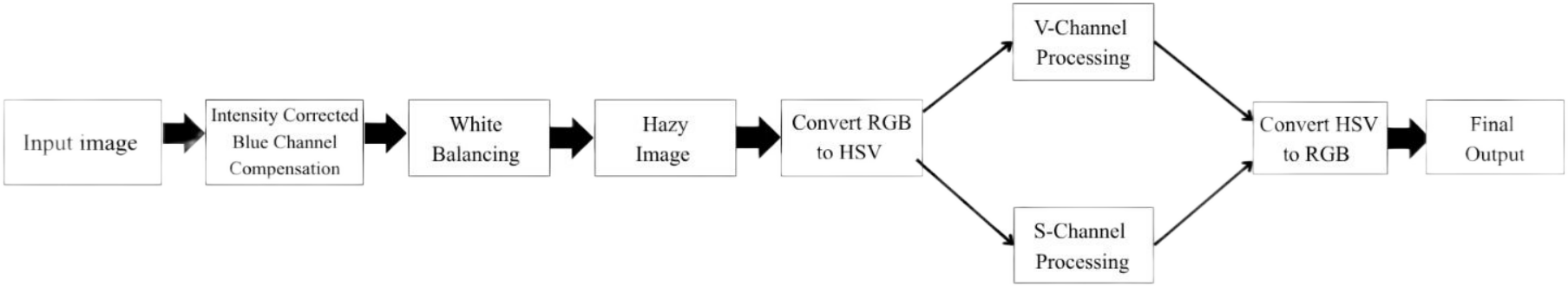

Fig.2: Existing Method Flow chart

A novel intensity-corrected blue channel compensation and edge-preserving contrast enhancement using laplace filter and sigmoid function for sand-dust image enhancement." IEEE Access 13 (2025): 43127-43144.

OBJECTIVES

- To develop an effective method for enhancing images degraded by fog, smoke, and sand dust conditions.
- To apply the **Dark Channel Prior (DCP)** technique for accurate haze estimation and removal.
- To estimate atmospheric light and transmission map for restoring scene radiance.
- To refine the transmission map for improved accuracy and smoothness in dehazed images.
- To enhance image contrast and visibility using **CLAHE** for better visual quality.
- To evaluate the performance of the proposed method using **PSNR** and **SSIM** metrics.

METHODOLOGY

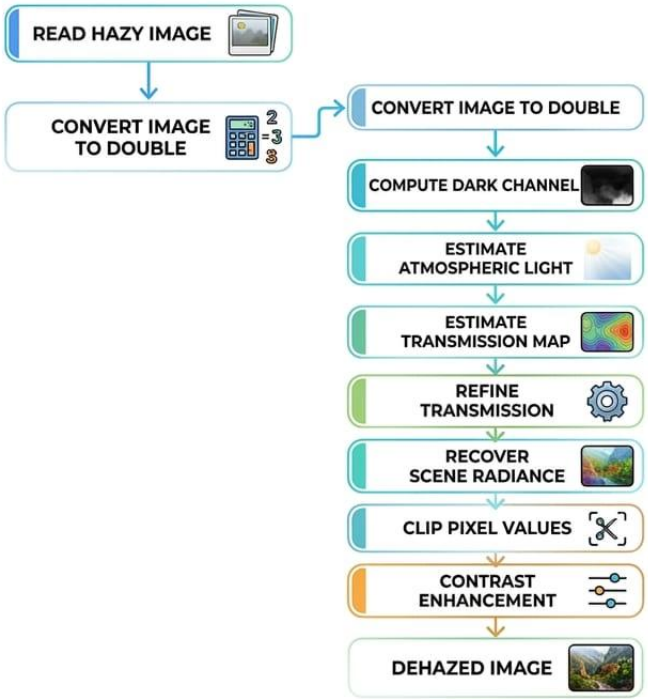

Fig.3: Image Dehazing Flow chart

PROPOSED WORK

- The proposed method performs image dehazing using the Dark Channel Prior (DCP) algorithm to remove fog from hazy images.
- The input hazy image is first converted to double precision and processed to compute the dark channel using a local patch.
- Atmospheric light is estimated from the brightest pixels in the dark channel to model the effect of haze.
- A transmission map is calculated and refined to estimate the amount of light reaching the camera through the fog.
- The scene radiance is recovered using the atmospheric scattering model, producing a clearer image.
- Finally, contrast enhancement and histogram analysis are applied to improve visibility and evaluate the dehazed result.

RESULTS – Qualitative Analysis

Fig 4: Different denoising and Enhanced methods on image 1

Fig.a: Input image 1

Fig.b: Ground Truth

Fig.c: Non linear filter image

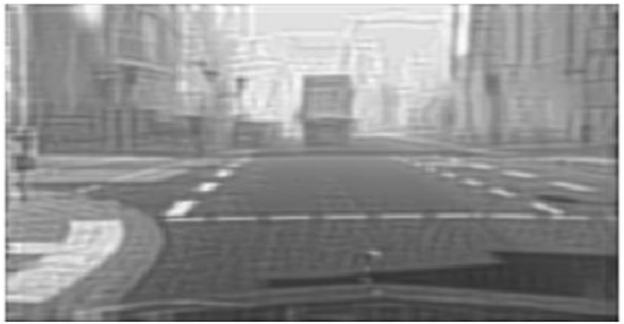

Fig.d: K-SVD+CLAHE image

Fig.e: DCP method

Fig 5: Different denoising and Enhanced methods on image 2

Fig.a: Input image 2

Fig.b: Ground Truth

Fig.c: Non linear filter image

Fig.d: K-SVD+CLAHE image

Fig.e: DCP method

Table 1: Quantitative analysis of hybrid combination of denoising and Enhanced methods on image 1

Method	PSNR (dB)	SSIM
Method 1(Non linear + clahe)	15.3567	0.5174
Method 2(K-SVD+clahe)	16.7938	0.5913
Method 3(DCP method)	**19.8245**	**0.7914**

Table 2: Quantitative analysis of hybrid combination of denoising and Enhanced methods on image 2

Method	PSNR (dB)	SSIM
Method 1(Non linear + clahe)	16.0353	0.5006
Method 2(K-SVD+clahe)	17.6712	0.5831
Method 3(DCP method)	**18.9542**	**0.7483**

Proposed method for Real time image

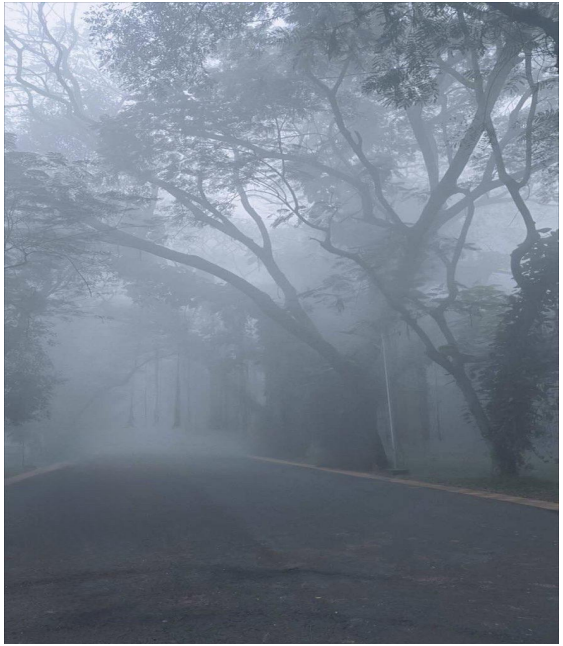

Fig 6a: Input image

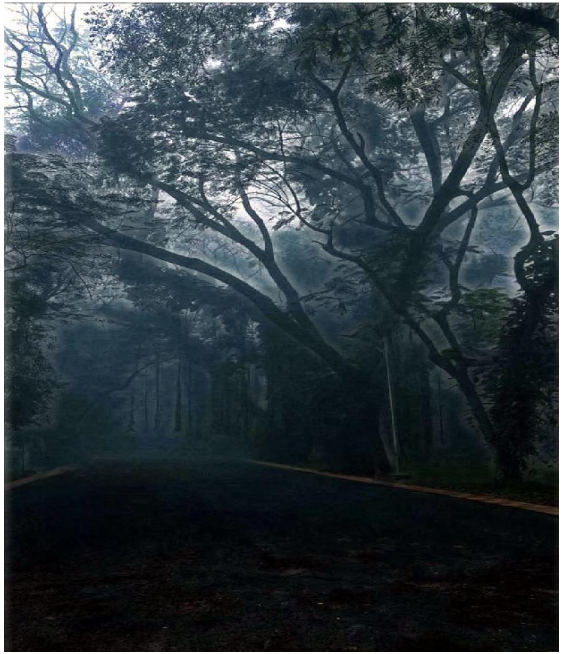

Fig 6b: Output image

CONCLUSION

- This work presents an effective image enhancement method for images degraded by fog, smoke, and sand dust.
- The combination of **Dark Channel Prior (DCP)** and **CLAHE** successfully improves visibility, contrast, and color quality.
- The methodaccurately estimates atmospheric light and transmission map to recover scene radiance.
- It preserves important image details and edges while removing haze effects.
- Experimental results show improved performance in terms of **PSNR** and **SSIM** compared to existing methods.
- Overall, the proposed approach produces clearer, more natural, and visually enhanced images suitable for real-world applications.
- The proposed method is simple, effective, and adaptable for different types of atmospheric degradations.

FUTURE SCOPE

- Integration of deep learning techniques for more robust and adaptive image enhancement.
- Extension of the method for real-time video dehazing applications.
- Improvement in handling extremely dense haze and complex atmospheric conditions.
- Application in advanced systems such as autonomous vehicles and drone-based monitoring.
- Combination with other enhancement techniques for further improvement in visual quality.

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Thank you